

# Complete Research Plan

## Deep Learning Subtype Discovery in Fibromyalgia via Emotion Regulation Task-fMRI

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### 1. Research Question

Fibromyalgia is clinically heterogeneous -- patients vary dramatically in pain severity, psychological comorbidity, emotional processing deficits, and treatment response. Current diagnosis is binary (FM vs healthy) and no imaging-based subtypes exist. This project asks: **can a self-supervised deep learning model trained on emotion regulation task-fMRI identify latent fibromyalgia subtypes with distinct neural and clinical signatures?**

This is the first application of deep learning to the task-fMRI modality in this dataset, and the first attempt at unsupervised neuroimaging subtyping in fibromyalgia using a publicly available open dataset.

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### 2. Dataset

**ds004144** -- available immediately at [openneuro.org/datasets/ds004144/versions/1.0.2](https://openneuro.org/datasets/ds004144/versions/1.0.2). No account, no approval, no email. Download today.

What you get:

- 33 fibromyalgia patients + 33 healthy female controls
  - T1w structural MRI, T2w, resting-state fMRI, task-fMRI
  - Task: emotion processing and regulation (Attend / Reappraise / Suppress) across positive, negative, and neutral visual stimuli -- 27.8 minutes, 834 volumes per participant
  - Clinical and behavioral data (pain scores, depression, anxiety, alexithymia, emotion regulation questionnaires) on Zenodo at [doi.org/10.5281/zenodo.6554870](https://doi.org/10.5281/zenodo.6554870)
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### 3. Technical Architecture

**Why self-supervised contrastive learning, not supervised classification:**

n=33 per group is too small to train a supervised deep learning classifier without severe overfitting. However contrastive self-supervised learning doesn't need labels -- it learns representations by maximizing similarity between augmented views of the same subject and

minimizing similarity between different subjects. This is the established solution for small-n neuroimaging.

### **Pipeline overview:**

#### **Step 1 -- Preprocessing**

Use fMRIPrep (containerized, runs locally) on the raw BIDS data. Outputs: motion-corrected, spatially normalized BOLD time series. Extract ROI time series using the Schaefer 200-parcel atlas -- this reduces dimensionality from voxels to 200 brain regions while preserving network structure. Compute functional connectivity matrices: 200x200 correlation matrices per subject per condition (Attend, Reappraise, Suppress x positive, negative, neutral = 9 condition matrices per subject).

#### **Step 2 -- Contrastive Self-Supervised Pretraining**

Build a graph neural network (GNN) encoder where each node is a brain region and edges are weighted by functional connectivity. Apply two augmentation strategies per subject to create positive pairs: (1) random ROI dropout (mask 10% of nodes), (2) Gaussian noise on edge weights. Train with NT-Xent contrastive loss -- same subject augmentations are pulled together, different subjects pushed apart. This produces a 64-dimensional latent embedding per subject per condition.

#### **Step 3 -- Subtype Discovery**

Pool embeddings across conditions using a simple attention-weighted average (learned weights per condition, interpretable). Apply UMAP for visualization and k-means clustering (k=2,3,4) on the pooled embeddings. Determine optimal k using silhouette score and gap statistic. Resulting clusters are your candidate FM subtypes.

#### **Step 4 -- Clinical Validation**

Test whether discovered subtypes differ on: pain intensity (VAS), depression (BDI), anxiety (BAI), alexithymia (TAS-20), emotion regulation (ERQ). Use Mann-Whitney U tests with FDR correction (small n, non-parametric). If subtypes show significantly different clinical profiles, you have a publishable finding.

#### **Step 5 -- Interpretability**

Use GNN attention weights to identify which brain regions and which task conditions (Attend vs Reappraise vs Suppress) drive subtype separation. Map back to functional networks (default mode, salience, executive control). This becomes your neuroscience interpretation section.

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## **4. Baselines**

You need baselines to show your deep learning approach adds value over simpler methods:

- K-means directly on raw FC matrices (no learned representation)
- PCA + k-means on FC matrices

- Traditional supervised SVM classification (FM vs HC) using FC features

Compare: do subtypes discovered by your contrastive model show better clinical separation than those from PCA+k-means? Do they show better within-cluster homogeneity?

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## 5. Week-by-Week Timeline

### Week 1 -- Setup and Preprocessing

Day 1: Download ds004144 and Zenodo clinical data. Verify BIDS structure. Set up conda environment: fmripreg, nilearn, torch, torch-geometric, umap-learn, sklearn, scipy, matplotlib, seaborn.

Day 2-3: Run fMRIPrep on all 66 subjects. This takes 6-12 hours on your RTX 5080 with Docker. While it runs, read the two papers that used this dataset to understand what's been done and frame your gap.

Day 4: Extract Schaefer 200 ROI time series from preprocessed outputs using nilearn. Compute 9 FC matrices per subject (3 conditions x 3 valences). Shape check: you should have 66 subjects x 9 conditions x 200 x 200 matrices.

Day 5: Exploratory analysis. Plot mean FC matrices per condition per group. Visualize group differences. Run basic statistical tests (t-tests on mean connectivity) to confirm the data shows signal. Load clinical data from Zenodo, merge with imaging data by subject ID.

Day 6-7: Build PyTorch Geometric dataset class. Each sample: a graph with 200 nodes, 200x200 edge weight matrix, subject label (FM/HC), condition label. Unit test -- verify shapes, no NaNs, correct subject-condition mapping.

### Week 2 -- Model Implementation and Training

Day 8: Implement GNN encoder. Use a 3-layer Graph Attention Network (GAT) from torch-geometric. Input: 200 node features (each node's time series mean, variance, and 3 spectral features = 5 features per node). Output: 64-dimensional graph-level embedding via global mean pooling.

Day 9: Implement augmentation pipeline. Two functions: random node masking (set 10% of node features to zero), edge noise (add Gaussian noise  $N(0, 0.1)$  to edge weights). For each training batch, generate two augmented views per graph and compute NT-Xent loss.

Day 10: Training loop. Train for 200 epochs with Adam optimizer ( $lr=1e-4$ ). Batch size 8 (all 66 subjects fit in ~2 GPU batches). Log loss curve. Expected behavior: loss decreases steeply for first 50 epochs then plateaus. If loss is NaN: reduce learning rate. If loss doesn't decrease: check augmentation isn't too aggressive.

Day 11: Debug day. Something will break -- budget the full day. Common issues: graph construction from FC matrix (threshold the matrix to keep top 20% of edges to sparsify), batch normalization blowing up with small batches (use layer norm instead), NT-Xent temperature parameter (start at 0.5).

Day 12: Extract embeddings from trained encoder for all 66 subjects x 9 conditions. Implement attention-weighted pooling across conditions. Run UMAP with  $n\_neighbors=10$ ,  $min\_dist=0.1$  on pooled embeddings. Plot colored by FM/HC label -- if your model is working, FM and HC should show some separation even though labels weren't used in training.

Day 13: K-means clustering ( $k=2,3,4$ ). Compute silhouette scores. Run baselines (PCA+k-means, raw FC k-means). Compare silhouette scores across methods. Statistical test: are your discovered subtypes significantly different from random clustering? Use permutation test (shuffle labels 1000 times, compute null silhouette distribution).

Day 14: Clinical validation. For each discovered subtype, compare pain scores, depression, anxiety, alexithymia using Mann-Whitney U. Apply FDR correction (Benjamini-Hochberg). This is the key result -- if subtypes differ clinically, you have a paper.

### **Week 3 -- Ablations, Interpretability, and Writing**

Day 15: Ablation studies. Train without augmentation (supervised contrastive with FM/HC labels). Train with only resting-state FC instead of task-fMRI. Train on individual conditions vs pooled. Compare subtype quality across ablations -- this shows the task-fMRI and contrastive learning both contribute.

Day 16: Interpretability. Extract GAT attention weights per node per subject. Average within subtypes. Plot brain maps showing which regions have highest attention weight in each subtype. Identify functional networks (use Schaefer network labels). Answer: does subtype 1 show elevated attention to salience network while subtype 2 shows elevated default mode? Map condition-specific attention weights: does Reappraise condition drive subtype separation more than Attend?

Day 17: Figure production. Build all 6 figures: (1) preprocessing pipeline diagram, (2) UMAP visualization of embeddings, (3) silhouette score comparison across methods, (4) clinical variable comparison between subtypes (violin plots), (5) brain map of attention weights per subtype, (6) condition-specific contribution to subtype separation.

Day 18: Write Methods and Results sections. Methods: preprocessing, architecture, training, evaluation. Results: model convergence, embedding quality, subtype discovery, clinical validation, interpretability. Be precise -- every number, every figure reference.

Day 19: Write Introduction and Discussion. Introduction structure: fibromyalgia heterogeneity problem (2 paragraphs), existing neuroimaging work and limitations (2 paragraphs), your specific gap and contribution (1 paragraph). Discussion: what do the subtypes mean clinically,

which brain networks drive them, limitations (small n, single site, female only, Mexican population), future directions (validation in larger cohort, treatment response prediction).

Day 20: Buffer. Whatever slipped goes here. If nothing slipped: polish figures, strengthen discussion, write abstract.

Day 21: Submit to journal. Target: Frontiers in Neuroscience (Pain and Neuroimaging section) or PLOS ONE. Both accept methodology-focused papers with small samples when the approach is novel and results are honestly framed. Write cover letter emphasizing: first deep learning application to task-fMRI in fibromyalgia, first unsupervised subtyping using neural data in this disease, fully open data and reproducible code.

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## 6. Software Stack

```
conda create -n fm-research python=3.11
conda install pytorch torchvision torchaudio pytorch-cuda=12.1 -c pytorch -c nvidia
pip install torch-geometric nilearn fmripred umap-learn scikit-learn scipy matplotlib seaborn
pandas nibabel wandb
```

fMRIPrep runs via Docker: `docker pull nipreps/fmripred:latest`

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## 7. Key Risks and Mitigations

**Small n:** Explicitly frame as proof-of-concept and exploratory. Use permutation testing throughout. Report effect sizes. Reviewers expect this for n=33 -- don't hide it, frame it.

**fMRIPrep failing on some subjects:** Budget for 10-15% subject dropout from motion or preprocessing failures. n=28 per group is still workable.

**No subtype separation found:** Still publishable as a negative result if the methodology is sound. Frame as "task-fMRI functional connectivity does not reveal robust subtypes in this sample, motivating larger studies." Frontiers accepts honest negative results.

**Clinical variables don't differ between subtypes:** Same framing -- negative result with methodological contribution. The contrastive learning approach itself is novel regardless of outcome.

**fMRI preprocessing taking too long:** fMRIPrep is parallelizable. Run 4 subjects simultaneously on your RTX 5080 + CPU cores. Total preprocessing time: ~20-30 hours spread across day 2-3.

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## 8. Publication Target

**Primary:** Frontiers in Neuroscience -- Pain and Neuroimaging section. Average time to first decision: 6-8 weeks. Open access. Accepts methodology papers with small n when framed correctly.

**Backup:** PLOS ONE -- accepts on methodological soundness, not impact. Guaranteed review if methods are correct.

**Novelty statement for cover letter:** "This is the first application of self-supervised contrastive learning to task-fMRI data in fibromyalgia, and the first data-driven neuroimaging subtype discovery in this condition using publicly available open data. Unlike prior work on this dataset which applied traditional GLM analysis to resting-state fMRI, we leverage the emotion regulation task paradigm -- which has not previously been analyzed with machine learning -- to discover latent patient subtypes with distinct clinical profiles."

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## 9. Code Release

Push all code to GitHub before submission. Repository structure:

```
fm-subtype-discovery/  
  preprocessing/  
    run_fmriprep.sh  
    extract_roi_timeseries.py  
    compute_fc_matrices.py  
  models/  
    gnn_encoder.py  
    contrastive_loss.py  
    augmentations.py  
  analysis/  
    clustering.py  
    clinical_validation.py  
    interpretability.py  
  notebooks/  
    01_exploratory_analysis.ipynb  
    02_training.ipynb  
    03_results.ipynb  
  README.md  
  requirements.txt
```

Reviewers increasingly check GitHub repos. A clean, documented repo strengthens your paper substantially.

